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Accuracy Test # 10

Based On NTA Abhyas (PRE-MEDICAL)

CAPACITIOR

1. A cylindrical capacitor has charge Q and length L. If both the charge and the length of the capacitor are doubled by keeping the other parameters fixed, then the energy stored in the capacitor

(1) Remains same

(2) Increases two times

(3) Decreases two times

(4) Increases four times

2. A capacitance of $2\mu F$ is required in an electrical circuit across a potential difference of 1.0 kV. A large number of $1\mu F$ capacitors are available which can withstand a potential difference of not more than 300V. The minimum number of capacitors required to achieve this is :

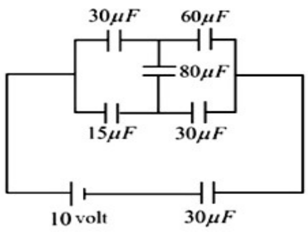
(1) 32

(2) 22

(3) 16

(4) 24

3. In the circuit shown below, the charge on 60 μF capacitor is



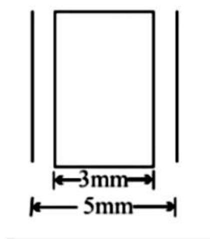
(1) 150 μC

(2) 100 μC

(3) 50 μC

(4) 75 μC

4. Separation between the plates of a parallel plate capacitor is 5mm. This capacitor, having air as the dielectric medium between the plates, is charged to a potential difference 25 V using a battery. The battery is then disconnected and a dielectric slab of thickness 3 mm and dielectric constant $\text{K} = 10$ is placed between the plates as shown. Potential difference between the plates after the dielectric slab has been introduced.



(1) 18.5 V

(2) 13.5 V

(3) 11.5 v

(4) 6.5 V

5. A capacitor of capacitance $100\mu F$ is charged by connecting it to a battery of EMF $12V$ and internal resistance 2Ω. The time taken before 99% of the maximum charge is stored on the capacitor.

(1) 0.92 ms

(2) 0.4 ms

(3) 0.8 ms

(4) 0.1 ms

6. A positive charge q is given to each plate of a parallel plate air capacitor having plate area A and plate separation d, then

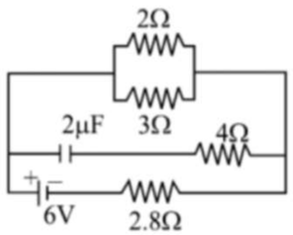
(1) since both the plates are identically charged, therefore , capacitance becomes equal to zero

(2) energy stored to in the space between the capacitor plate is equal to $\frac{q^2}{\varepsilon_0 A^2}$

(3) no charge appears on the inner surface of the plates

(4) potential difference between the plates is equal to $\frac{2qd}{\varepsilon_0 A}$

7. In the figure shown, the capacity of the condenser C is $2\mu F$. The current in 2Ω resistor after steady state is -



(1) 9 A

(2) 0.9 A

(3) $\frac{1}{9}$ A

(4) $\frac{1}{0.9}$ A

8. A dielectric slab is inserted between the plates of an isolated capacitor. The force between the plates will

(1) increase

(2) decrease

(3) remain unchanged

(4) become zero

9. The effective capacitance of two capacitors of capacitances C_1 and C_2 (C_2> C_1) connected in parallel is 25/6 times the effective capacitance when they are connected in series. The ratio $\frac{C_2}{C_1}$ is –

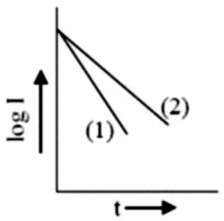
(1) 3/2

(2) 4/3

(3) 5/3

(4) 25/6

10. A capacitor of capacitance C is charged to a potential difference V and then connected in series with an open key and a pure resistor R. At time t = 0, the key is closed. If I is current at time t = 0, a plot of logI against t is shown in the graph (2). Later one of the parameters, i.e., V, R and C is changed, keeping the other two constant and graph (2) is recorded. Then



(1) C is reduced

(2) C is increased

(3) R is reduced

(4) R is increased

11. Three capacitors each of 4 ,μF are to be connected in such a way that the effective capacitance is 6 μF. This can be done by connecting them

(1) all in series

(2) all in parallel

(3) two in parallel and one in series

(4) two in series and one in parallel

12. Two condensers of capacities 2C and C are joined in parallel and charged up to potential V. The battery is removed and the condenser of capacity C is filled completely with a medium of dielectric constant K. The potential difference across the capacitors will now be

(1) $\frac{3V}{K+2}$

(2) $\frac{3V}{K}$

(3) $\frac{V}{K+2}$

(4) $\frac{V}{K}$

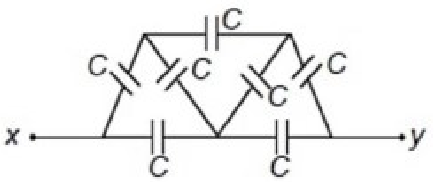
13. Two capacitor of capacitance 6 μF and 12 μF are connected in series with a battery. The voltage across the 6 μF capacitor is 2 V. the total battery voltage is

(1) 4 V
(2) 6 V
(3) 9 V
(4) 3 V

14. A parallel plate capacitor of capacitance C is connected to a battery and is charged to a potential difference V .another capacitor of capacitance 2C is similarly charged to a potential difference 2V. the charging battery is now disconnected and the capacitors are connected in parallel to each other in such a way that the positive terminal of one is connected to the negative terminal of other. the final energy of the configuration is

(1) Zero
(2) $\frac{25}{6}CV^2$
(3) $\frac{9}{2}CV^2$
(4) $\frac{3}{2}CV^2$

15. Equivalent capacitance between x and y is



(1) $\frac{7}{8}C$
(2) $\frac{8}{7}C$
(3) $\frac{7}{9}C$
(4) $\frac{9}{7}C$

16. If the potential of a capacitor having capacity 6μF is increased from 10V to 20V,then increase in its energy is

(1) 12×10^{-6} J
(2) 9×10^{-4} J
(3) 4.5×10^{-6} J
(4) 2.25×10^{-4} J

17. How does the electric field (E) between the plates of a charged cylindrical capacitor vary with the distance r from the axis of the cylinder?

(1) $E \propto \frac{1}{r^2}$
(2) $E \propto \frac{1}{r}$
(3) $E \propto r^2$
(4) $E \propto r$

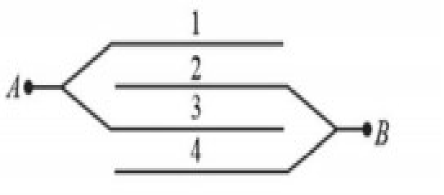
18. A $4\mu F$ capacitor is charged to 400 V and then its plates are joined through a resistance of 1 kΩ. The heat produced in the resistance is

(1) 0.16 J
(2) 1.28 J
(3) 0.64 J
(4) 0.32 J

19. The capacity of an isolated sphere is increased n times when it is enclosed by an earthed concentric sphere. The ratio of their radii is

(1) $\frac{n^2}{n-1}$
(2) $\frac{n}{n-1}$
(3) $\frac{2n}{n+1}$
(4) $\frac{2n+1}{n+1}$

20. Four metal plates numbered 1, 2, 3 and 4 are arranged, as shown in figure. The area of each plate is A and the separation between them is d. The capacitance of the arrangement is



(1) $\frac{\varepsilon_0 A}{d}$

(2) $\frac{2\varepsilon_0 A}{d}$

(3) $\frac{3\varepsilon_0 A}{d}$

(4) $\frac{4\varepsilon_0 A}{d}$

21. The two plates of a parallel plate capacitor are 4 mm apart. A slab of dielectric constant 3 and thickness 3mm is introduced between the plates with its faces parallel to them. The distance between the plates is so adjusted that the capacitance of the capacitor becomes 2/3 rd of its original value. What is the new distance between the plates?

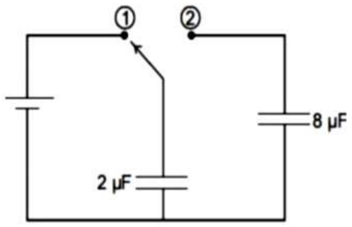
(1) 9mm

(2) 21mm

(3) 5mm

(4) 8mm

22. A 2 μF capacitor is charged as shown in the figure. The percentage of its stored energy dissipated as heat after the switch S is turned to position 2 is



(1) 0%

(2) 20%

(3) 75%

(4) 80%

23. Two insulated charged spheres of radii R_1 and R_2 having charges Q_1 and Q_2, respectively, are connected to each other; then there is

(1) no change in the energy of the system

(2) an increase in the energy of the system

(3) always a decrease in the energy of the system

(4) a decrease in energy of the system unless $q_1R_2 = q_2R_1$

24. A condenser of capacitance $10\mu F$ has been charged to 100V. It is now connected to another uncharged condenser in parallel. The common potential becomes 40 V. The capacitance of another condenser is

(1) 15 μF

(2) 5 μF

(3) 10 μF

(4) 16 μF

25. A parallel plate capacitor is charged by a battery after charging the capacitor, the battery is disconnected and if a dielectric plate is inserted between the plates, then which of the following statements is not correct.

(1) increase in the stored energy

(2) decrease in the potential difference

(3) decrease in the electric field

(4) increase in the capacitance

26. The capacitance of a parallel plate condenser does not depend upon

(1) the distance between the plates

(2) area of the plates

(3) medium between the plates

(4) metal of the plates

27. Two identical metal plates are given positive charges Q_1 and $Q_2 (Q_2 < Q_1)$respectively. If they are now brought close together to form a parallel plate capacitor with capacitance C, the potential difference between them is

(1) $\frac{Q_1+Q_2}{2C}$

(2) $\frac{Q_1+Q_2}{C}$

(3) $\frac{Q_1-Q_2}{C}$

(4) $\frac{Q_1-Q_2}{2C}$

28. What is the largest charge a metal ball of 1 mm radius can hold?
(Dielectric strength of air is 3×10^6 Vm^{-1}).

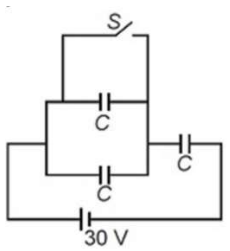
(1) 3 nC

(2) $\frac{1}{3}$ nC

(3) 2 nC

(4) $\frac{1}{2}$ nC

29. Three capacitors each of capacitance C = 2μF are connected with a battery of emf 30 V as shown in the figure. When the switch S is closed, the heat generated in the circuit will be



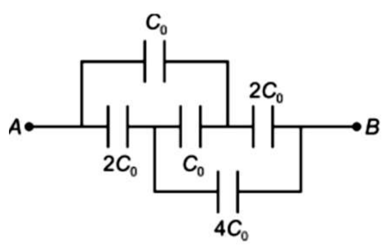
(1) 0.15 mJ

(2) 0.30 mJ

(3) 0.10 mj

(4) 0.60 mJ

30. The equivalent capacitance between the points A and B is



(1) $2C_0$

(2) C_0

(3) $4C_0$

(4) 6 C_0

31. The separation between the plates of a parallel plate capacitor, connected to a battery (zero resistance) of constant EMF is increased with constant (very slow) speed by external forces. During the process, w is the work done by external forces. ΔU Is the change in potential energy of the capacitor w_b is work done by the battery and H is the heat loss in the circuit. Then

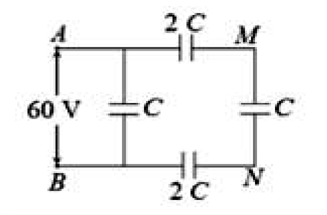
(1) $w + w_b = \Delta U$

(2) $H \neq 0$

(3) $H = \Delta U$

(4) $w = 0$

32. In the circuit shown, a potential difference of 60 V is applied across AB. The potential difference between the points M and N is



(1) 10V

(2) 15V

(3) 20V

(4) 30V

33. Two identical capacitors have the same capacitance C. One of them is charged to potential V_1 and other to V_2. If their like-charged plates are then connected together then the decrease in energy of the combined system is

(1) $\frac{1}{4}C(V_1^2 - V_2^2)$

(2) $\frac{1}{4}C(V_1^2 + V_2^2)$

(3) $\frac{1}{4}C(V_1 + V_2)^2$

(4) $\frac{1}{4}C(V_1 - V_2)^2$

34. A 40 μF capacitor in a defibrillator is charged to 3000 V. The energy stored in the capacitor is sent through the patient during a pulse of duration 2 ms. The power delivered to the patient is

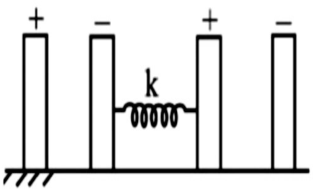
(1) 45 kW

(2) 90 kW

(3) 180 kW

(4) 360 kW

35. Two charged capacitors have their outer plates fixed and inner plates connected by a spring of force constant k, the magnitude of charge on each capacitors is q and sign of charge is shown in figure. Find the extension in the spring at equilibrium.



(1) $\frac{q^2}{2A\varepsilon_0 K}$

(2) $\frac{q^2}{4A\varepsilon_0 K}$

(3) $\frac{q^2}{A\varepsilon_0 K}$

(4) zero

36. Capacitor C_1 of capacity 1 μF and capacitor C_2 of capacity 2 μF are separately charged fully by a common battery. The two capacitors are then separately allowed to discharge through equal resistors at t = 0

(1) At t = 0 the value of current in the circuit containing 1 μF is more than current in the circuit containing 2 μF

(2) At t = 0 the current in 2 μF capacitor circuit is more than current in 1 μF capacitor circuit

(3) 1 μF capacitor losses 50 % charge sooner than 2 μF capacitor

(4) 2 μF capacitor losses 50 % charge sooner than 1 μF capacitor

37. A fully charged capacitor has a capacitance C. it is discharged through a small coil of resistance wire embedded in a thermally insulated block of specific heat capacity s and mass m. if the temperature of the block is raised by ΔT, the potential difference V across the capacitance is

(1) $\sqrt{\frac{2mC\Delta T}{s}}$

(2) $\frac{mC\Delta T}{s}$

(3) $\frac{ms\Delta T}{C}$

(4) $\sqrt{\frac{2ms\Delta T}{C}}$

38. If an electron enters into a space between the plates of parallel plate capacitor at an angle ∝ with the plates and leaves at an angle β to the plates .The ratio of its kinetic energy while entering the capacitor to that while leaving will be

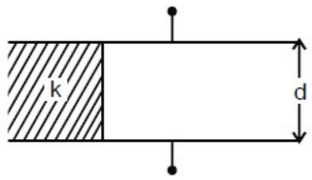
(1) $\left(\frac{\sin\beta}{\sin\alpha}\right)^2$

(2) $\left(\frac{\cos\beta}{\cos\alpha}\right)^2$

(3) $\left(\frac{\cos\alpha}{\cos\beta}\right)^2$

(4) $\left(\frac{\sin\alpha}{\sin\beta}\right)^2$

39. A parallel plate capacitor with air as a dielectric has capacitance C. A slab of dielectric constant K having same thickness as the separation between the plates is introduced so as to fill one fourth of the capacitor as shown in the figure. The new capacitance will be



(1) $(K+3)\frac{C}{4}$

(2) $(K+2)\frac{C}{4}$

(3) $(K+1)\frac{C}{4}$

(4) $(K)\frac{C}{4}$

40. Two identical capacitors each of capacitance 5μF are charged to potentials 2kV and 1kV respectively. Their - ve ends are connected together. When the +ve ends are also connected together, the loss of energy of the system is

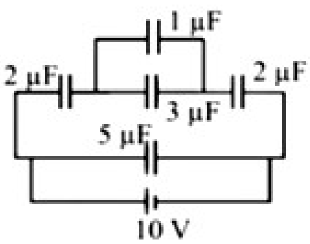
(1) 160 J

(2) Zero

(3) 5 J

(4) 1.25 J

41. The ratio of potential differences between $1\mu F$ and $5\mu F$ capacitors is



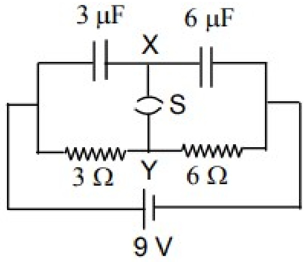
(1) 1:2

(2) 3:1

(3) 1:5

(4) 10:1

42. A circuit is connected as shown in the figure with switch S is open. When the switch is closed, the total amount of charge that flows from X to Y is



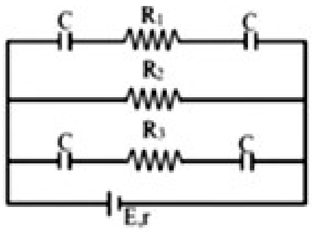
(1) 0

(2) 54 μC

(3) 81 μC

(4) 27μC

43. In the adjoining circuit diagram, E=5V, r =1 Ω, R_2=4 Ω, R_1=R_3=1 Ω and C=3μF. Then the numerical value of the charge on each plate of the capacitor is



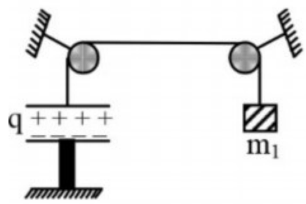
(1) 24 μC

(2) 12μC

(3) 6μC

(4) 3μC

44. In the given system a capacitor of plate area A is charged up to charge q. The mass of each plate is m_2 .The lower plate is rigidly fixed. Find the value of m_1, so that the system is in equilibrium –



(1) $m_2 + \frac{q^2}{\varepsilon_0 A g}$

(2) m_2

(3) $\frac{q^2}{2A\varepsilon_0 g} + m_2$

(4) None of these

45. The plates of a parallel plate air capacitor are charged to 100 V. A 2mm plate is inserted between the plates of the capacitor. To maintain the same potential difference , the distance between the capacitor plates is increased by 1.6mm. The dielectric constant of the plate is

(1) 1.25

(2) 5

(3) 2.5

(4) 4